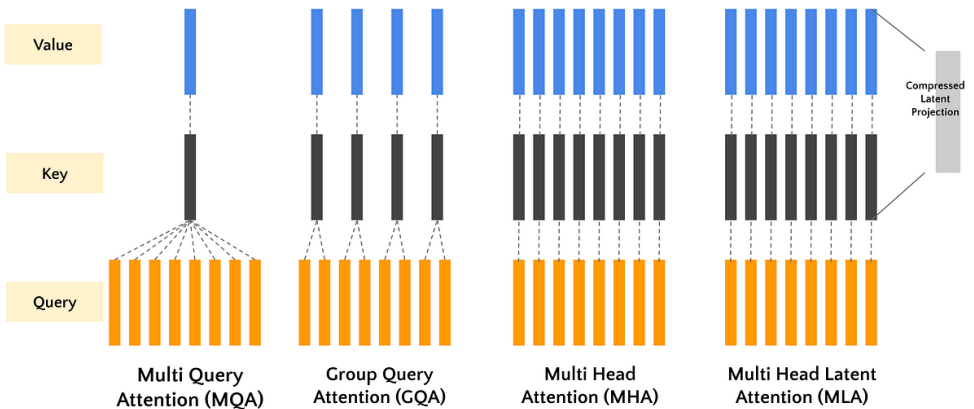


# Attention Variants Explained

## Attention – MQA | GQA | MHA | MLA



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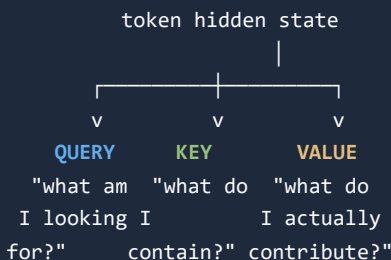
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# Quick Refresher: Q, K, V

Every token produces three vectors. Everything here is about rearranging these three things.



## HOW THEY COMBINE

$$\text{Attention}(Q, K, V) = \text{softmax} \left( \frac{Q \cdot K^T}{\sqrt{d}} \right) \cdot V$$

— match queries against keys —

— weighted sum —



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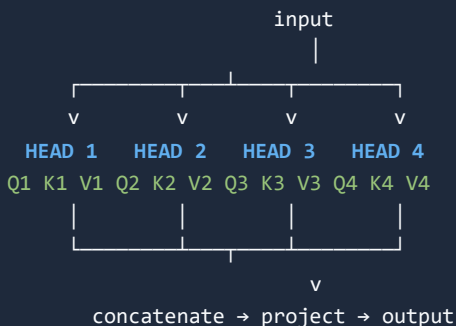
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# Multi-Head Attention: The Original

Instead of one computation, run several **heads** in parallel. Each head has its own independent Q, K, and V projections.



- ✓ **Why:** Lets the model track grammar, coreference, and position simultaneously.

## THE COUNTING THAT MATTERS LATER

32 heads = 32 Q sets, 32 K sets, 32 V sets



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# The Problem: KV Cache Memory

KV cache size  $\approx 2 \times \text{layers} \times \text{heads} \times \text{dim} \times \text{seq\_len} \times \text{batch}$

▲                      ▲                      ▲

K,V                  grows!                grows!

Model weights are a fixed cost. The KV cache is *per request*. On a serving GPU, the KV cache determines how many users you can handle concurrently.

## Weights (Fixed)

KV Cache (Grows per user)



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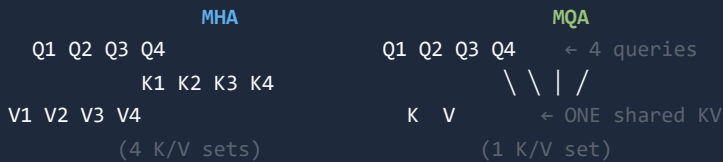


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# Multi-Query Attention: Share K and V

**The insight:** queries differentiate the heads. Do keys and values need to be separate? Keep many queries, share a single K and V across all heads.



## ↓ THE PAYOFF

For 32 heads, MQA reduces cache size by ~32x. More concurrent users, faster decoding.

## ⚠ THE COST

Quality degrades measurably. Heads lose their ability to specialize independently.



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# Grouped-Query Attention: The Compromise

GQA sits between the two extremes: split heads into groups, and give each group its own shared K and V.

**MHA**

|    |    |    |    |
|----|----|----|----|
| Q1 | Q2 | Q3 | Q4 |
| K1 | K2 | K3 | K4 |
| V1 | V2 | V3 | V4 |

4 K/V sets  
best quality  
largest cache

**GQA (2 groups)**

|     |     |     |     |
|-----|-----|-----|-----|
| Q1  | Q2  | Q3  | Q4  |
| \   | /   | \   | /   |
| K_a | V_a | K_b | V_b |

2 K/V sets  
near-MHA quality  
much smaller cache

**MQA**

|    |    |    |    |
|----|----|----|----|
| Q1 | Q2 | Q3 | Q4 |
| \  | \  |    | /  |
|    |    | K  | V  |

1 K/V set  
lowest quality  
smallest cache

**Why this is the modern default:** GQA recovers nearly all of MHA's quality while capturing most of MQA's memory savings.

## READING A MODEL CONFIG

num\_attention\_heads: 32 ← query heads  
num\_key\_value\_heads: 8 ← K/V groups  
→ GQA with 4 query heads per group



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# MHA vs MQA vs GQA: One dial, three settings

MHA ————— GQA ————— MQA  
(K/V per head) (K/V per group) (one K/V total)

QUALITY



CACHE



SPEED



HIGHEST QUALITY

BEST BALANCE ★

FASTEST / SMALLEST

All three compute *identical math*. They differ only in how many K/V sets exist. A model with a 128K+ context window almost certainly uses GQA or MQA.



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# Flash Attention: A Different Kind of Fix

MQA and GQA change the architecture. Flash Attention changes **how the computation executes on the GPU**. The math and results are exactly the same.

## THE PROBLEM: MEMORY TRAFFIC

1. compute  $Q \cdot K^T$  → write full  $n \times n$  matrix to slow GPU memory
2. read it back → apply softmax → write result back
3. read it back → multiply by  $V$  → write output

For 4,000 tokens: a  $4,000 \times 4,000$  matrix (16M values) is shuttled back and forth repeatedly.

Attention is **memory-bandwidth-bound**. The GPU spends its time waiting on data movement, not doing arithmetic.

## THE BOTTLENECK

SRAM (fast, tiny)



HBM (large, slow)



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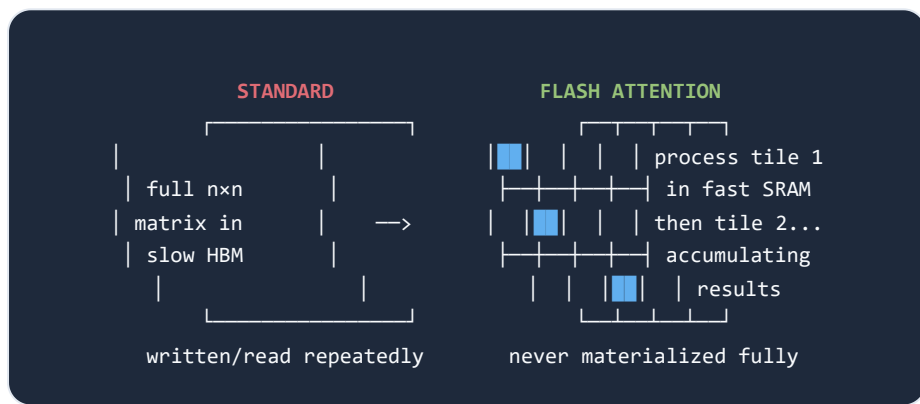
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# How Flash Attention Works:

## Never write the big matrix

Flash Attention breaks the computation into **tiles** small enough to fit in the GPU's fast SRAM, computing softmax incrementally.



- ✓ **Online softmax:** computes softmax incrementally across tiles.
- ✓ **Identical outputs:** this is NOT an approximation.
- ✓ **Linear memory scaling:** instead of quadratic.



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## DISTINCTION

# Flash Attention Is Not An architecture change

| MQA / GQA                | FLASH ATTENTION                        |
|--------------------------|----------------------------------------|
| Changes the model itself | Changes the GPU kernel                 |
| Different weights        | Same weights, untouched                |
| Decided at training time | Swap in at any time                    |
| Affects output quality   | <b>Output is bit-for-bit identical</b> |
| A design tradeoff        | <b>A free win, no tradeoff</b>         |

### THEY ARE ORTHOGONAL

ARCHITECTURE  
MHA/MQA/GQA

+

KERNEL  
Flash Attention

Used TOGETHER, not either/or



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## SUMMARY

# All Four Compared: The full picture

|              | MHA          | MQA             | GQA                 | Flash Attention |
|--------------|--------------|-----------------|---------------------|-----------------|
| Type         | Architecture | Architecture    | Architecture        | Kernel optim.   |
| K/V sets     | One per head | One total       | One per group       | Unchanged       |
| Output       | -            | Yes (some loss) | Yes (minimal)       | No (identical)  |
| Cache size   | Largest      | Smallest        | Small               | Unchanged       |
| Main benefit | Best quality | Max memory      | Balanced            | Speed + memory  |
| Main cost    | Cache size   | Quality drop    | Slight quality drop | Needs hardware  |
| Set at       | Training     | Training        | Training            | Any time        |



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# Where This Shows Up For You: Reading configs

```
# Spotting the variant in a model config
config.num_attention_heads    # 32 ← query heads
config.num_key_value_heads    # 8  ← K/V groups

# equal          → MHA
# second is 1    → MQA
# in between     → GQA (here: 4 query heads per group)

# Enabling Flash Attention
model = AutoModelForCausalLM.from_pretrained(
    "some-model",
    attn_implementation="flash_attention_2",
    torch_dtype=torch.bfloat16, # required for FA2
)
```

- ✓ Long context is possible because of GQA .
- ✓ Concurrency depends on GQA/MQA .
- ✓ Flash Attention is almost free , enable it.
- ❗ PagedAttention (vLLM) is separate; manages blocks.



## ✖ MISCONCEPTIONS

# Common Misconceptions: What people get wrong

### ✖ ~~Flash = approximation~~

False. Mathematically identical results. It changes scheduling, not math.

### ✖ ~~GQA hurts quality a lot~~

Overstated. It lands very close to MHA, which is why it's the default.

### ✖ ~~These fix quadratic cost~~

False for MQA/GQA (shrinks cache). Flash makes memory linear, compute is still quadratic.

### ✖ ~~Chosen at inference time~~

False for MHA/MQA/GQA (baked in at training). Only Flash can be swapped later.

### ✖ ~~MQA is strictly better~~

False. The quality cost is real, exactly why GQA displaced it.

### ✖ ~~Flash vs GQA is either/or~~

False. They're orthogonal and almost always used together.



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# Two problems. Two fixes.



**ARCHITECTURE:** MHA → MQA → GQA

(shrinks the KV cache)



**KERNEL:** Flash Attention

(cuts memory traffic, free)

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